

CONTACT INFORMATION	Department of Mathematics Indian Institute of Technology Jodhpur N.H. 62, Nagaur Road, Karwar Jodhpur - 342030, Rajasthan, India	Research Coordinates:    Email: kumar.131[at]iitj.ac.in pankaj.iitj95[at]gmail.com
RESEARCH INTEREST	Distribution Theory, Copulas, Risk Theory, Change Point Analysis, Statistical Modeling, ML/DL, Spatial Statistics,Real Data Analysis	
EDUCATION	Ph.D. in Mathematics Specialization in Statistics and Finance Indian Institute of Technology Jodhpur Advisor(s): Dr. V V M S Chandramouli and Dr. Vivek Vijay Thesis Title: On Statistical Distribution Theory and Copula with Application to Financial Risk Measures	July 2021 – October 2026 CGPA : 8.42 Thesis Submitted: April 2026
	M.Tech. in Data and Computational Sciences Indian Institute of Technology Jodhpur Advisor: Dr. Vivek Vijay Thesis Title: Machine Learning Models for Stock Selection Based on Fundamental Ratios and Optimal Portfolio Selection	July 2019 – May 2021 CGPA : 7.82
	Integrated MSc in Mathematics Pondicherry University, Puducherry, India Minor in: Computer Science and Statistics	July 2013 – May 2018 CGPA : 7.19
EXPERIENCES	Research Consultant WorldQuant BRAIN India, WorldQuant, LLC	November 2023 – January 2025
	Summer Intern Exposys Data Labs,Bengaluru – Predicting Diabetes Data With Machine Learning Techniques	June 2021 – July 2021
	Projet Intern @ Centre for Mathematical and Computational Economics Indian Institute of Technology Jodhpur – Time Series Analysis for economic data using machine learning tools	March 2021 – May 2021
TECHNICAL SKILLS	Programming ML Tools Software Tools	R, Python, Matlab PyTorch, TensorFlow, Pandas, Numpy, Scikit-learn RStudio, Jupyter Notebook, L^AT_EX

Publications

[1]

Kumar, P. (2026). Dependence Analysis and Financial Risk Estimations: A Hybrid Approach. *Finance Research Open*, 2 (3), 100147.

[2]

Kumar, P., & Vijay, V. (2026). A Copula-Based Bivariate q -Weibull Distribution. *Lobachevskii Journal of Mathematics*, 47 (1), 435–448.

[3]

Kumar, P. and Vijay, V. (2026). A heavy-tail arctan-based mixture model for modeling financial data. *International Journal of Information Technology* (Accepted)

[4]

Kumar, P. & Vijay, V. (2027) Estimation of VaR and CVaR using Regular Vine Copula and GJR-GARCH model for intra-sectoral portfolio. *Finance: Theory and Practice*. 31 (1).

[5]

Kumar, P., Mandal, B., & Vijay, V. (2026). Entropy-based trivariate q -Gaussian model (TV q -GD): an application to financial data. *European Journal of Statistics*, 6,14-14.

[6]

Kumar, P., & Vijay, V. (2026). Arctan Alpha Power Transformation Family of Distributions: Properties, Estimation and Application. (In Review)

[7]

Kumar, P. & Vijay, V. (2026). Use of GJR-GARCH model for forecasting of financial risk. *Proceedings of The Eleventh International Conference on Mathematics and Computing* (pp. 503–512). Singapore: Springer Nature Singapore.

- [8] **Kumar, P., & Vijay, V. (2024).** Analyzing dependencies in the electric vehicle sector using the Canonical Vine Copula. *Proceedings of the 6th International Conference on Statistics: Theory and Applications*, vol. 168, pp. 1–6.
- [9] **Kumar, P., & Vijay, V. (2025).** Dependence Analysis of Indian Energy Sector Using Regular Vine Copula and Information Transfer Entropy. *Integrated Uncertainty in Knowledge Modelling and Decision Making* (pp. 174-185). Singapore: Springer Nature Singapore.
- [10] **Mandal, B., Kumar, P. & Vijay, V. (2026).** Entropic Value at Risk (EVaR) for Heavy-Tailed Distributions and Backtesting. *Proceedings of the Twelfth International Conference on Mathematics and Computing*. (Accepted)
- [11] **Kumar, P. (2026).** Change-point detection of the generalised Gaussian model based on the modified information criterion. (Working Paper)
- [12] **Kumar, P. (2026).** Forecasting the entropic value at risk under the variance gamma model. (Working Paper)

Conference Talks and Workshop Participation

CONTRIBUTED TALKS

- [1] Title: “Dependence analysis of indian energy sector using regular vine copula and information transfer entropy”, presented at the **11th International Symposium on Integrated Uncertainty in Knowledge Modelling and Decision Making (IUKM 2025)**, Vietnam (17 - 19 March 2025).
- [2] Title: “Use of gjr-garch model for forecasting of financial risk”, presented at the **11th International Conference on Mathematics and Computing (ICMC 2025)**, IIT Bhilai (09 - 11 January 2025).
- [3] Title: “Analyzing dependencies in the electric vehicle sector using the canonical vine copula”, presented at the **6th International Conference on Statistics: Theory and Applications (ICSTA’24)**, Barcelona, Spain (19 - 21 August 2024).
- [4] Title: “Stock selection using fundamental ratios”, presented at the **23rd International Conference on Economic Modeling and Data Science (EcoMod 2024)**, Çesme, Turkey (03 - 05 July 2024).

POSTER PRESENTED

- [1] Title: “EU-EV and vine copula approach for intra-sectoral dependence in indian stock market”, poster presented at the **9th Conference on Statistical Methods in Finance (StatFin 2024)**, CMI and ISI, Chennai (19 - 21 December 2024).

CONFERENCE ATTENDED

- [1] **26th ACM Conference on Economics and Computation (EC’25)**, Stanford University, Stanford, CA, USA (07 - 10 July 2025)
- [2] **11th National Conference on Technology and Innovation in Mathematics Education (TIME)**, hosted by math4all and IIT Jodhpur (07-09 December 2023).
- [3] **Research Symposium on Finance and Economics (RSFE)**, Krea University, India (12 - 14 June 2024)

WORKSHOP PARTICIPATION

- [1] **2026 Penn Causal Inference Summer Institute Workshop**, Organized by University of Pennsylvania, USA (1 - 5 June 2026).
- [2] **Workshop on Data Science: Probabilistic and Optimization Methods**, Organized by ICTS Bengaluru (4 - 15 August 2025).
- [3] **Workshop on Mathematical Finance and Risk Management**, Organized by IIT Madras (15 - 16 February 2025).
- [4] **9th Workshop on Statistical Methods in Finance (StatFin 2024)**, hosted by CMI and ISI, Chennai (17 - 18 December 2024)
- [5] **Large Scale Optimization Workshop**, Organized by Indian Institute of Management (IIM) Ahmedabad (5 - 13 May 2022).
- [5] **International workshop On Intelligent Data Analysis Techniques**, Organized by India-Taiwan Cooperation (2015) at Pondicherry University (17 March 2015)

Awards, Grants & Achievements

AWARDS, GRANTS, & ACHIEVEMENTS	The Penn Causal Inference Summer Institute Travel Scholarship, University of Pennsylvania, USA	2026
	American Statistical Association (ASA) Travel Grant for SDSS26, Milwaukee, USA	2026
	ACM SIGTravel Grant for EC25 at Stanford University, USA	2025
	Institute Travel Grant for StatFin24 at CMI and ISI, Chennai and ICMC25, IIT Bhilai	2025
	WorldQuant Challenge Gold Certificate in Quantitative Finance Hackathon	2023
	Received Senior Research Fellowship (SRF) from Ministry of Education (MoE) for PH.D.	2023 – 2026
	Received Junior Research Fellowship (JRF) from Ministry of Education (MoE) for PH.D.	2021 – 2023
	Received Ministry of Education (MoE) Scholarship for Postgraduate Education	2019 – 2021
	Qualified Graduate Aptitude Test in Engineering (GATE) in Mathematics	2019
	Qualified Joint Admission test for Masters (JAM)	2016
	Received Merit Cum Means Scholarship from UGC for Undergraduate Education	2013 – 2016
	Qualified Joint Entrance Examination Main (JEE Main)	2013

Professional Services

REVIEWER	Risk and Decision Analysis (SAGE Publishing)	
VOLUNTEER	ACM Student Chapter IIT Jodhpur	2023
	Symposium on Advanced Mathematical Modelling and Computing (SAMMC)	2023
	11th National Conference on Technology and Innovation in Mathematics Education (TIME)	2023
MEMBERSHIPS	Bachelier Finance Society (BFS)	2026
	Institute of Mathematical Statistics (IMS)	2026 - 2027
	International Indian Statistical Association (IISA)	2026 - 2027
	Royal Statistical Society (RSS)	2023 - 2026

Mentorship & Teaching

MENTORSHIP EXPERIENCE	Collaborated with students on multiple Bachelor’s and Master’s thesis projects, IIT Jodhpur	2021–2025
TEACHING ASSISTANT EXPERIENCE	Helped teach various graduate and undergraduate level courses by designing and evaluating exams, assignments, and projects, conducted tutorial and lab sessions.	2021 — 2025
Tutorial Courses: Mathematics-1, Mathematics-2, Probability and Statistics, Linear Models, Financial Engineering, Optimization for Data Science, Probability, Statistics and Stochastic Processes, Introduction to Financial Engineering, Statistics for Data Science, Optimization for Machine Learning, Financial Analytics, Foundations of Statistics and Probability		
Laboratory Courses (R and Python): Mathematics-2, Data Analytics, Scientific Computations, Mathematical Modeling and Simulations, Optimization for Machine Learning,		
SUBJECT MATTER EXPERT (SME)	Subject Matter Expert in Mathematics, Chegg, Inc.	2017 – 2022

REFERENCES

Dr. V. V. M. S Chandramouli (Email: chsarma@iitj.ac.in)
Associate Professor, Department of Mathematics, IIT Jodhpur

Dr. Vivek Vijay (Email: vivekvijayvargiya@gmail.com)

Dr. Kirankumar R Hiremath (Email: k.r.hiremath@iitj.ac.in)
Professor, Department of Mathematics, IIT Jodhpur

Dr. Sandeep Kumar Yadav (Email: sy@iitj.ac.in)
Professor, Department of Electrical Engineering, IIT Jodhpur